

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

Sec:- 3

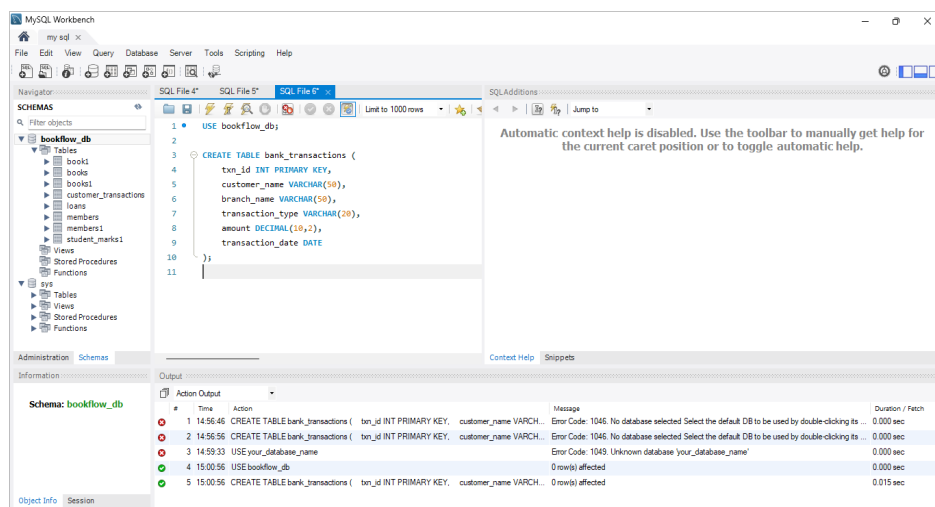
Name:- Srithika Kambhammettu

Roll no:- 2520030361

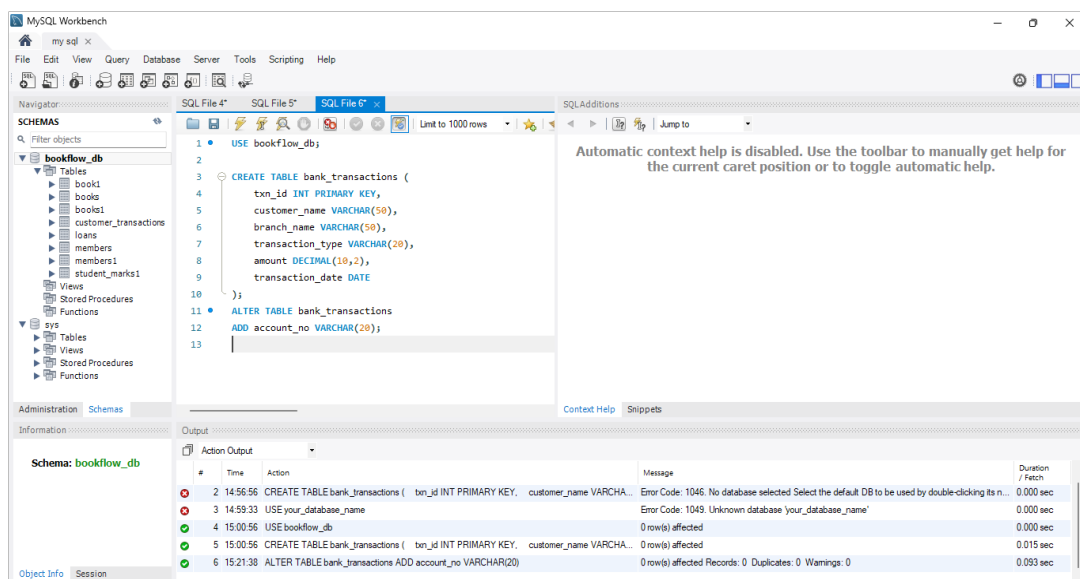
1- DDL, DML, DCL, TCL

SECTION 1 : DDL (DATA DEFINITION LANGUAGE)

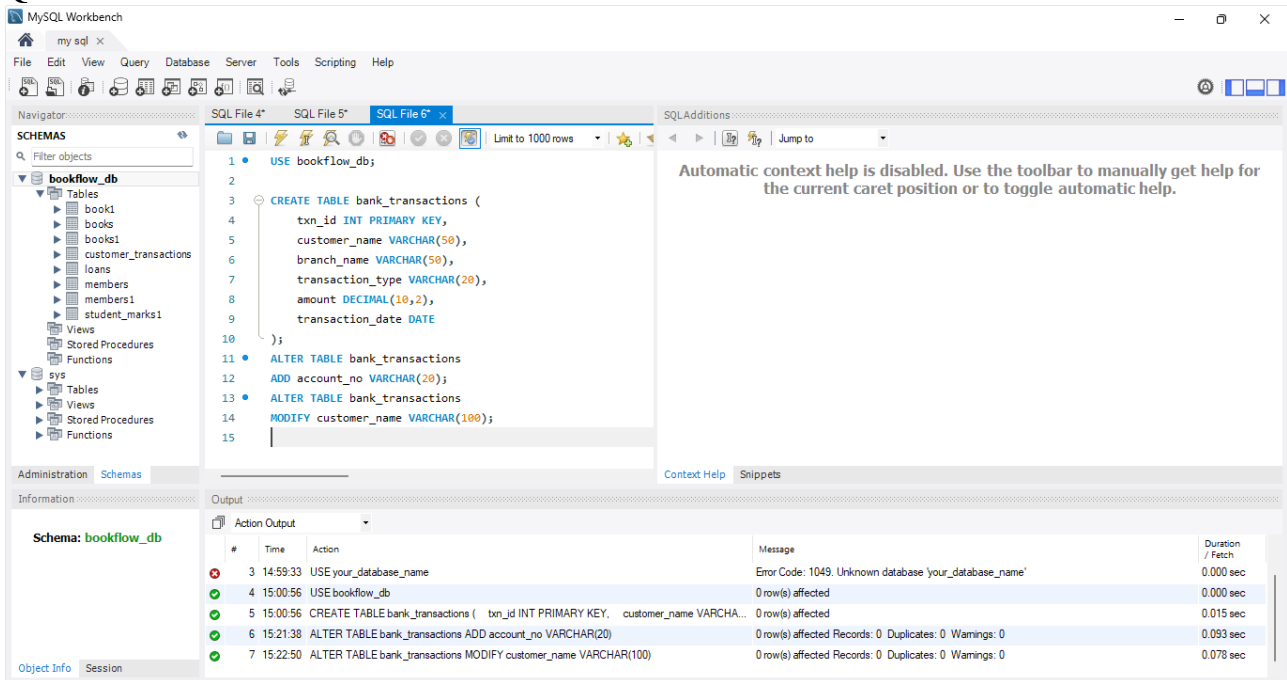
QUESTION 1 - CREATE TABLE



QUESTION 2 - ALTER TABLE ADD COLUMN



QUESTION 3 - ALTER TABLE MODIFY COLUMN



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator: my sql x

SQL File 4* SQL File 5* SQL File 6* x

Limit to 1000 rows

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

Schema: bookflow_db

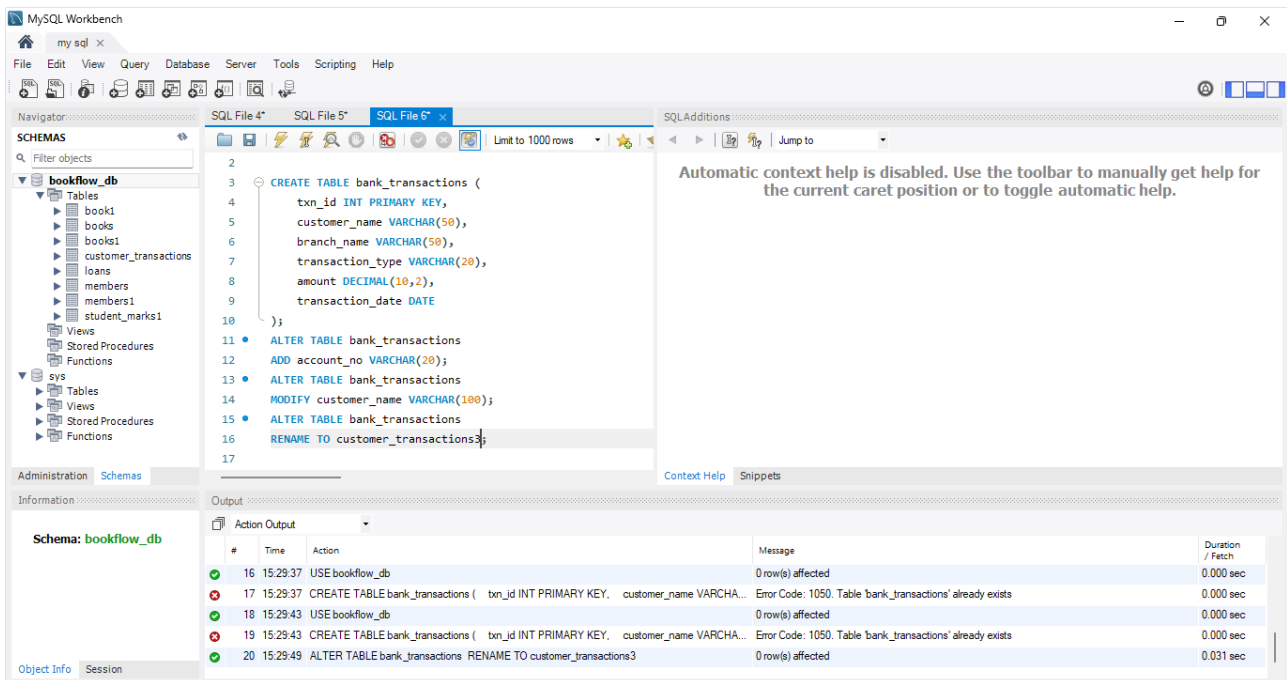
Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	14:59:33	USE your_database_name	Error Code: 1049. Unknown database 'your_database_name'	0.000 sec
4	15:00:56	USE bookflow_db	0 row(s) affected	0.000 sec
5	15:00:56	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name VARCHAR(50), transaction_type VARCHAR(20), amount DECIMAL(10,2), transaction_date DATE	0 row(s) affected	0.015 sec
6	15:21:38	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.093 sec
7	15:22:50	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec

QUESTION 4 - RENAME TABLE



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator: my sql x

SQL File 4* SQL File 5* SQL File 6* x

Limit to 1000 rows

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

Schema: bookflow_db

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
16	15:29:37	USE bookflow_db	0 row(s) affected	0.000 sec
17	15:29:37	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name VARCHAR(50), transaction_type VARCHAR(20), amount DECIMAL(10,2), transaction_date DATE	Error Code: 1050. Table 'bank_transactions' already exists	0.000 sec
18	15:29:43	USE bookflow_db	0 row(s) affected	0.000 sec
19	15:29:43	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name VARCHAR(50), transaction_type VARCHAR(20), amount DECIMAL(10,2), transaction_date DATE	Error Code: 1050. Table 'bank_transactions' already exists	0.000 sec
20	15:29:49	ALTER TABLE bank_transactions RENAME TO customer_transactions3	0 row(s) affected	0.031 sec

QUESTION 5 - TRUNCATE TABLE

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator: my sql x

SQL File 4* SQL File 5* SQL File 6*

Limit to 1000 rows

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

SCHEMAS

Filter objects

bookflow_db

Tables

bank_transactions

book1

books

books1

customer_transactions

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	14:35:55	SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject HA...	1 row(s) returned	0.016 sec / 0.000 sec
2	14:36:47	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.047 sec

QUESTION 6 - DROP TABLE

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator: my sql x

SQL File 4* SQL File 5* SQL File 6*

Limit to 1000 rows

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

SCHEMAS

Filter objects

bookflow_db

Tables

bank_transactions

book1

books

books1

customer_transactions

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	14:45:44	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.032 sec
7	14:46:40	CREATE TABLE bank_backup (id INT)	0 row(s) affected	0.031 sec
8	14:46:51	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.031 sec
9	14:47:29	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
10	14:47:43	CREATE TABLE bank_backup (id INT)	0 row(s) affected	0.031 sec
11	14:48:49	DROP TABLE bank_backup	0 row(s) affected	0.015 sec

Query Completed

SECTION 2 : DML (DATA MANIPULATION LANGUAGE)

STEP 1 : INSERT SAMPLE DATA

MySQL Workbench interface showing a query execution session. The query window displays the following SQL code:

```
29 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
30 SELECT * FROM customer_transactions;
31 UPDATE customer_transactions
32 SET customer_name = 'Srithika'
33 WHERE txn_id = 101;
34 SELECT * FROM customer_transactions;
35
```

The Result Grid shows the following data:

txn_id	customer_name	branch_name	transaction_type	amount	t
101	Srithika	Hyderabad	Deposit	5000.00	2
102	Sita	Hyderabad	Withdrawal	2000.00	2
103	Kiran	Vijayawada	Deposit	12000.00	2
104	Anil	Vizag	Deposit	8000.00	2
105	Priya	Hyderabad	Withdrawal	3500.00	2
106	Ramesh	Vizag	Deposit	15000.00	2

The Output pane shows the following execution results:

#	Time	Action	Message	Duration / Fetch
18	15:12:22	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	Error Code: 1146. Table 'bookflow_db.customer_transactions1' doesn't exist	0.000 sec
19	15:12:30	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	Error Code: 1062. Duplicate entry '101' for key 'customer_transactions.PRIMARY'	0.016 sec
20	15:12:37	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	Error Code: 1062. Duplicate entry '101' for key 'customer_transactions.PRIMARY'	0.000 sec
21	15:13:13	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
22	15:14:35	UPDATE customer_transactions SET customer_name = 'Srithika' WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
23	15:14:49	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

QUESTION 7 – INSERT

MySQL Workbench interface showing a query execution session. The query window displays the following SQL code:

```
28 (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
29 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
30 SELECT * FROM customer_transactions;
31 UPDATE customer_transactions
32 SET customer_name = 'Srithika'
33 WHERE txn_id = 101;
34 SELECT * FROM customer_transactions;
35
36 INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...
37 VALUES
38 (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
39
40
41
42
```

The Output pane shows the following execution results:

#	Time	Action	Message	Duration / Fetch
22	15:14:35	UPDATE customer_transactions SET customer_name = 'Srithika' WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
23	15:14:49	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
24	15:15:46	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18)	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
25	15:15:49	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18)	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
26	15:16:14	INSERT INTO customer_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18)	Error Code: 1136. Column count doesn't match value count at row 1	0.000 sec
27	15:18:03	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	1 row(s) affected	0.000 sec

QUESTION 8 – UPDATE

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 4* SQL File 5* SQL File 6*

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
24	15:15:46	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
25	15:15:49	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
26	15:16:14	INSERT INTO customer_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	Error Code: 1136. Column count doesn't match value count at row 1	0.000 sec
27	15:18:03	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	1 row(s) affected	0.000 sec
28	15:20:24	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
29	15:20:42	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec

Query Completed

QUESTION 9 – DELETE

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 4* SQL File 5* SQL File 6*

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

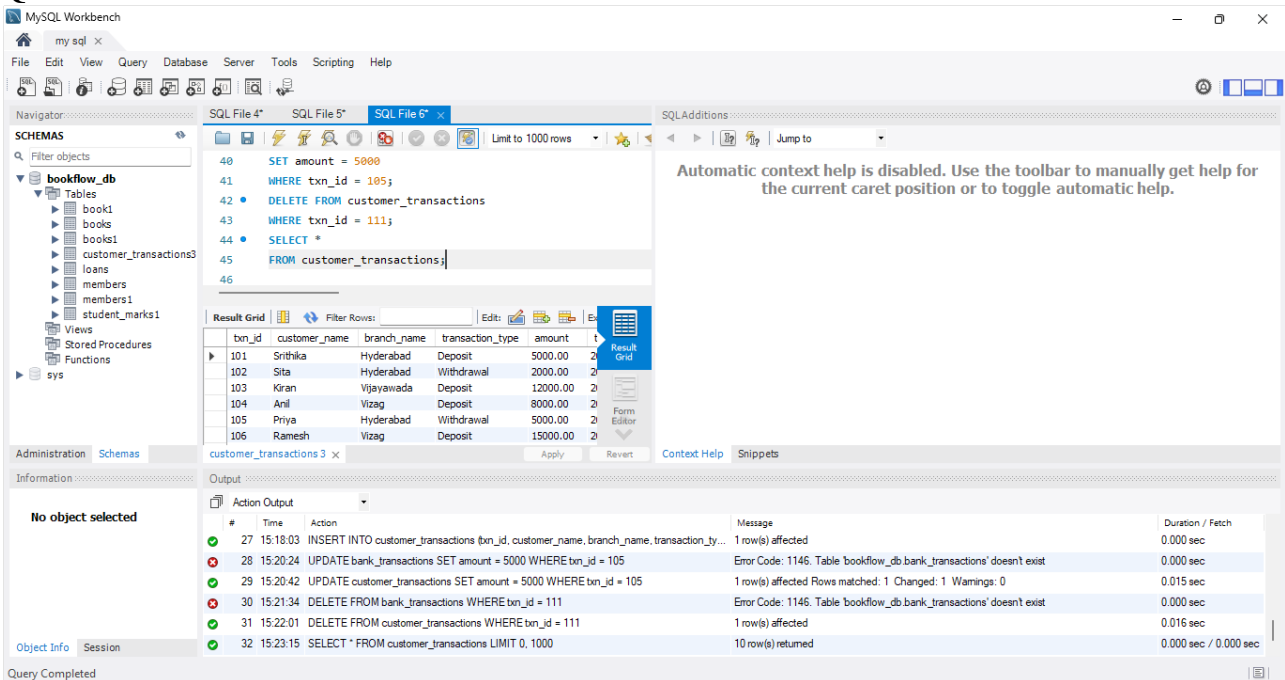
Output

Action Output

#	Time	Action	Message	Duration / Fetch
26	15:16:14	INSERT INTO customer_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	Error Code: 1136. Column count doesn't match value count at row 1	0.000 sec
27	15:18:03	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	1 row(s) affected	0.000 sec
28	15:20:24	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
29	15:20:42	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
30	15:21:34	DELETE FROM bank_transactions WHERE txn_id = 111	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
31	15:22:01	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec

Query Completed

QUESTION 10 – SELECT



MySQL Workbench interface showing a SQL query in SQL File 4. The query is:

```
40 SET amount = 5000
41 WHERE txn_id = 105;
42 DELETE FROM customer_transactions
43 WHERE txn_id = 111;
44 SELECT *
45 FROM customer_transactions;
46
```

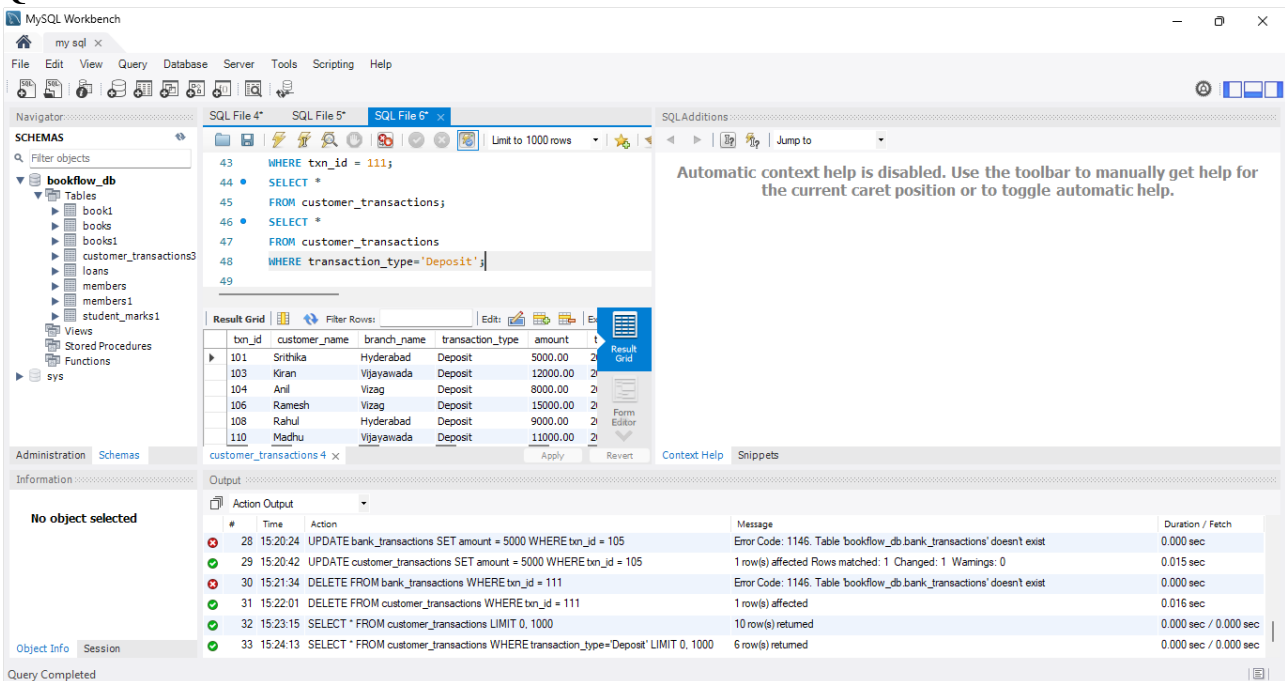
The Result Grid shows the following data:

txn_id	customer_name	branch_name	transaction_type	amount	t
101	Srithika	Hyderabad	Deposit	5000.00	2
102	Sita	Hyderabad	Withdrawal	2000.00	2
103	Kiran	Vijayawada	Deposit	12000.00	2
104	Anil	Vizag	Deposit	8000.00	2
105	Priya	Hyderabad	Withdrawal	5000.00	2
106	Ramesh	Vizag	Deposit	15000.00	2

The Output pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
27	15:18:03	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_ty...	1 row(s) affected	0.000 sec
28	15:20:24	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
29	15:20:42	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
30	15:21:34	DELETE FROM bank_transactions WHERE txn_id = 111	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
31	15:22:01	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec
32	15:23:15	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

QUESTION 11 - SELECT WITH WHERE



MySQL Workbench interface showing a SQL query in SQL File 4. The query is:

```
43 WHERE txn_id = 111;
44 SELECT *
45 FROM customer_transactions;
46 SELECT *
47 FROM customer_transactions
48 WHERE transaction_type='Deposit';
49
```

The Result Grid shows the following data:

txn_id	customer_name	branch_name	transaction_type	amount	t
101	Srithika	Hyderabad	Deposit	5000.00	2
103	Kiran	Vijayawada	Deposit	12000.00	2
104	Anil	Vizag	Deposit	8000.00	2
106	Ramesh	Vizag	Deposit	15000.00	2
108	Rahul	Hyderabad	Deposit	9000.00	2
110	Madhu	Vijayawada	Deposit	11000.00	2

The Output pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
28	15:20:24	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
29	15:20:42	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
30	15:21:34	DELETE FROM bank_transactions WHERE txn_id = 111	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
31	15:22:01	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec
32	15:23:15	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
33	15:24:13	SELECT * FROM customer_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

QUESTION 12 - SELECT WITH ORDER BY

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```
46 SELECT *
47 FROM customer_transactions
48 WHERE transaction_type='Deposit';
49 SELECT *
50 FROM customer_transactions
51 ORDER BY amount DESC;
52
```

The Results window displays the output of the second query, showing a list of transactions ordered by amount in descending order:

txn_id	customer_name	branch_name	transaction_type	amount	t
106	Ramesh	Vizag	Deposit	15000.00	2
103	Kiran	Vijayawada	Deposit	12000.00	2
110	Madhu	Vijayawada	Deposit	11000.00	2
108	Rahul	Hyderabad	Deposit	9000.00	2
104	Anil	Vizag	Deposit	8000.00	2
101	Srithika	Hyderabad	Deposit	5000.00	2

The Output window shows the execution log:

#	Time	Action	Message	Duration / Fetch
29	15:20:42	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
30	15:21:34	DELETE FROM bank_transactions WHERE txn_id = 111	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
31	15:22:01	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec
32	15:23:15	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
33	15:24:13	SELECT * FROM customer_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
34	15:25:26	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

SECTION 3 : DCL (DATA CONTROL LANGUAGE)

QUESTION 13 - GRANT SELECT

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```
44 SELECT *
45 FROM customer_transactions;
46 SELECT *
47 FROM customer_transactions
48 WHERE transaction_type='Deposit';
49 SELECT *
50 FROM customer_transactions
51 ORDER BY amount DESC;
52 CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
53 GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost';
54
55
56
57
58
```

The Results window displays the output of the last query, showing a list of transactions ordered by amount in descending order:

txn_id	customer_name	branch_name	transaction_type	amount	t
106	Ramesh	Vizag	Deposit	15000.00	2
103	Kiran	Vijayawada	Deposit	12000.00	2
110	Madhu	Vijayawada	Deposit	11000.00	2
108	Rahul	Hyderabad	Deposit	9000.00	2
104	Anil	Vizag	Deposit	8000.00	2
101	Srithika	Hyderabad	Deposit	5000.00	2

The Output window shows the execution log:

#	Time	Action	Message	Duration / Fetch
35	15:26:58	GRANT SELECT ON bank_transactions TO Auditor1	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.016 sec
36	15:27:09	GRANT SELECT ON customer_transactions TO Auditor1	Error Code: 1410. You are not allowed to create a user with GRANT	0.016 sec
37	15:29:07	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123'	0 row(s) affected	0.031 sec
38	15:29:55	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for 'Auditor1'@'localhost'	0.000 sec
39	15:30:55	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
40	10:15:06	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec

QUESTION 14 - GRANT ALL PRIVILEGES

MySQL Workbench interface showing the execution of SQL queries. The left sidebar displays the database schema for 'bookflow_db'. The main editor shows the following SQL queries:

```
44 SELECT *
45 FROM customer_transactions;
46 SELECT *
47 FROM customer_transactions
48 WHERE transaction_type='Deposit';
49 SELECT *
50 FROM customer_transactions
51 ORDER BY amount DESC;
52 CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
53 GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost';
54 CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456';
55 GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost';
```

The output window shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
38	15:29:55	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for 'Auditor1'@'localhost'	0.000 sec
39	15:30:55	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
40	10:15:06	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
41	10:16:07	GRANT ALL ON customer_transactions TO BranchManager	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
42	10:19:22	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456'	0 row(s) affected	0.015 sec
43	10:19:22	GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec

QUESTION 15 - REVOKE SELECT

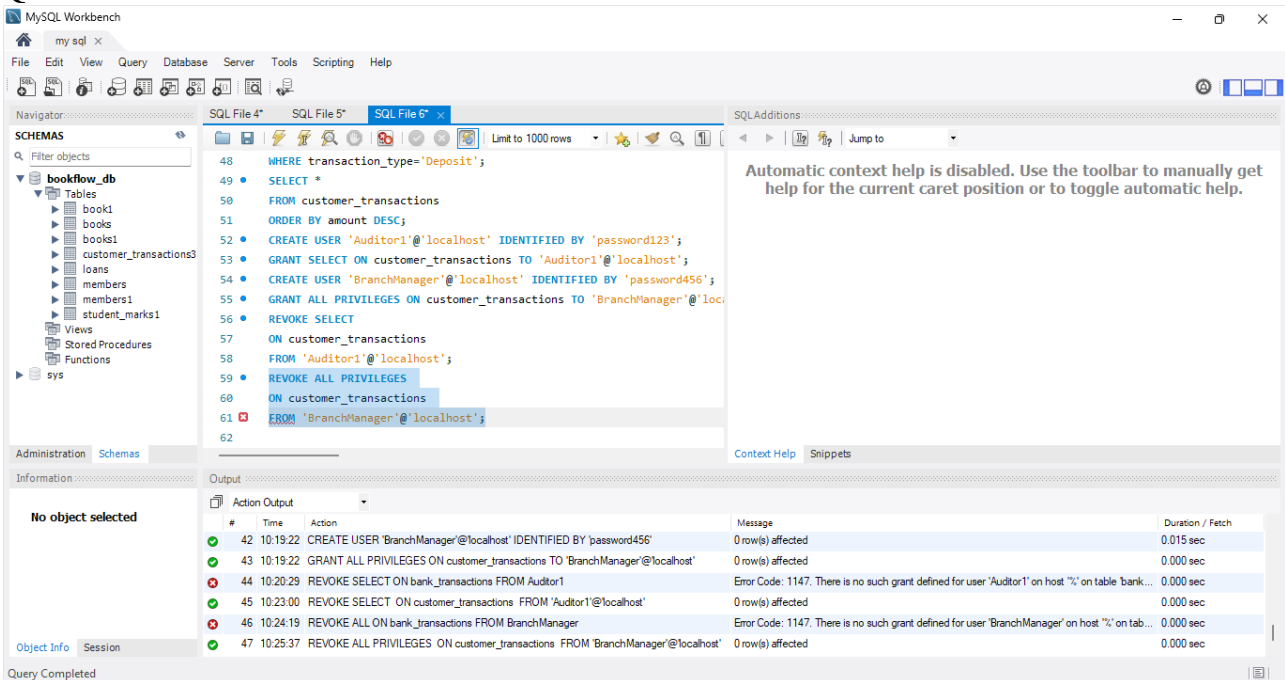
MySQL Workbench interface showing the execution of SQL queries. The left sidebar displays the database schema for 'bookflow_db'. The main editor shows the following SQL queries:

```
45 FROM customer_transactions;
46 SELECT *
47 FROM customer_transactions
48 WHERE transaction_type='Deposit';
49 SELECT *
50 FROM customer_transactions
51 ORDER BY amount DESC;
52 CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
53 GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost';
54 CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456';
55 GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost';
56 REVOKE SELECT
57 ON customer_transactions
58 FROM 'Auditor1'@'localhost';
```

The output window shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
40	10:15:06	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
41	10:16:07	GRANT ALL ON customer_transactions TO BranchManager	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
42	10:19:22	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456'	0 row(s) affected	0.015 sec
43	10:19:22	GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
44	10:20:29	REVOKE SELECT ON bank_transactions FROM Auditor1	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_...	0.000 sec
45	10:23:00	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec

QUESTION 16 - REVOKE ALL



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window shows a SQL script with the following content:

```
48 WHERE transaction_type='Deposit';
49 SELECT *
50 FROM customer_transactions
51 ORDER BY amount DESC;
52 CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
53 GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost';
54 CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456';
55 GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost';
56 REVOKE SELECT
57 ON customer_transactions
58 FROM 'Auditor1'@'localhost';
59 REVOKE ALL PRIVILEGES
60 ON customer_transactions
61 FROM 'BranchManager'@'localhost';
62
```

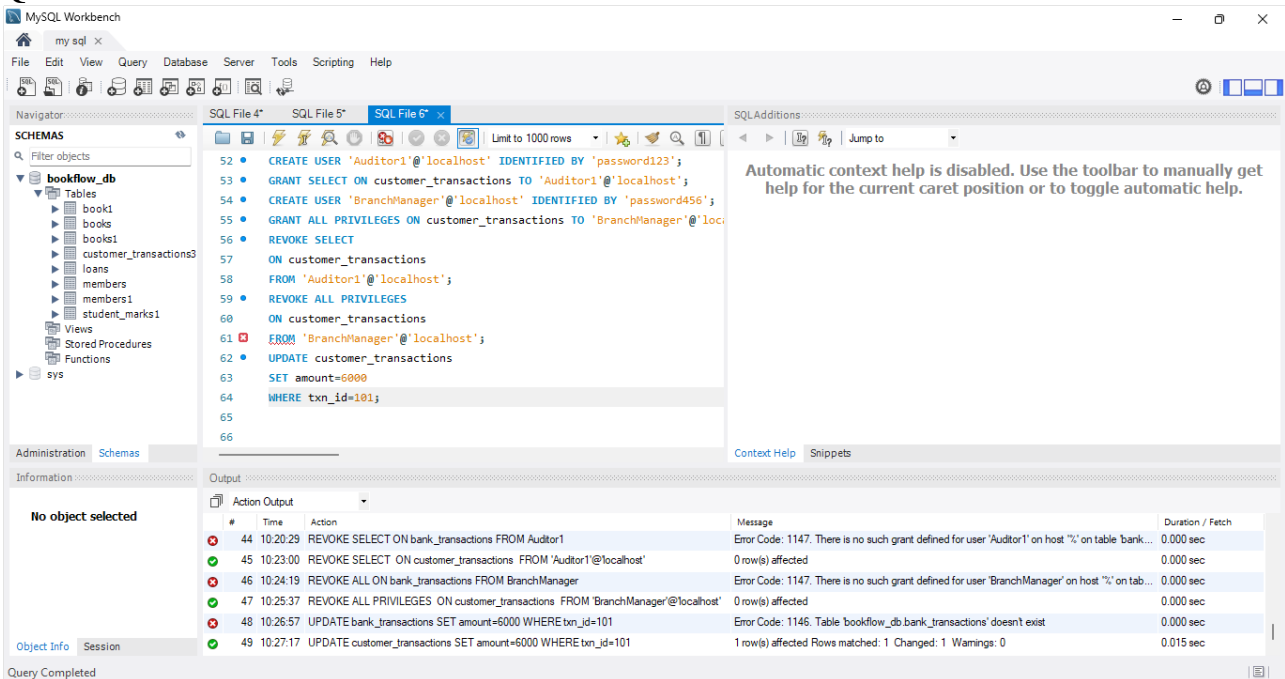
The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
42	10:19:22	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456'	0 row(s) affected	0.015 sec
43	10:19:22	GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
44	10:20:29	REVOKE SELECT ON bank_transactions FROM Auditor1	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host '%' on table 'bank...	0.000 sec
45	10:23:00	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
46	10:24:19	REVOKE ALL ON bank_transactions FROM BranchManager	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host '%' on tab...	0.000 sec
47	10:25:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec

Query Completed

SECTION 4 : TCL (TRANSACTION CONTROL LANGUAGE)

QUESTION 17 – COMMIT



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window shows a SQL script with the following content:

```
52 CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
53 GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost';
54 CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456';
55 GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost';
56 REVOKE SELECT
57 ON customer_transactions
58 FROM 'Auditor1'@'localhost';
59 REVOKE ALL PRIVILEGES
60 ON customer_transactions
61 FROM 'BranchManager'@'localhost';
62 UPDATE customer_transactions
63 SET amount=6000
64 WHERE txn_id=101;
65
66
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
44	10:20:29	REVOKE SELECT ON bank_transactions FROM Auditor1	Error Code: 1147. There is no such grant defined for user 'Auditor1' on host '%' on table 'bank...	0.000 sec
45	10:23:00	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
46	10:24:19	REVOKE ALL ON bank_transactions FROM BranchManager	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host '%' on tab...	0.000 sec
47	10:25:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
48	10:26:57	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
49	10:27:17	UPDATE customer_transactions SET amount=6000 WHERE txn_id=101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec

Query Completed

QUESTION 18 – SAVEPOINT

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
52 • CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
53 • GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost';
54 • CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456';
55 • GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost';
56 • REVOKE SELECT
57   ON customer_transactions
58   FROM 'Auditor1'@'localhost';
59 • REVOKE ALL PRIVILEGES
60   ON customer_transactions
61   FROM 'BranchManager'@'localhost';
62 • UPDATE customer_transactions
63   SET amount=6000
64   WHERE txn_id=101;
65 • SAVEPOINT Before_Update;
66
```

#	Time	Action	Message	Duration / Fetch
45	10:23:00	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
46	10:24:19	REVOKE ALL ON bank_transactions FROM BranchManager	Error Code: 1147. There is no such grant defined for user 'BranchManager' on host '%' on tab...	0.000 sec
47	10:25:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM BranchManager@'localhost'	0 row(s) affected	0.000 sec
48	10:26:57	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
49	10:27:17	UPDATE customer_transactions SET amount=6000 WHERE txn_id=101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
50	10:27:58	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec

QUESTION 19 - ROLLBACK TO SAVEPOINT

the toolbar to manually get help for the current caret position or to toggle automatic help.

```
54 • CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password456';
55 • GRANT ALL PRIVILEGES ON customer_transactions TO 'BranchManager'@'localhost';
56 • REVOKE SELECT
57   ON customer_transactions
58   FROM 'Auditor1'@'localhost';
59 • REVOKE ALL PRIVILEGES
60   ON customer_transactions
61   FROM 'BranchManager'@'localhost';
62 • UPDATE customer_transactions
63   SET amount=6000
64   WHERE txn_id=101;
65 • SAVEPOINT Before_Update;
66 • UPDATE customer_transactions
67   SET amount=99999
68   WHERE txn_id=102;

```

#	Time	Action	Message	Duration / Fetch
47	10:25:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM BranchManager@'localhost'	0 row(s) affected	0.000 sec
48	10:26:57	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
49	10:27:17	UPDATE customer_transactions SET amount=6000 WHERE txn_id=101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
50	10:27:58	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
51	10:28:36	UPDATE bank_transactions SET amount=99999 WHERE txn_id=102	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
52	10:28:50	UPDATE customer_transactions SET amount=99999 WHERE txn_id=102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec

QUESTION 20 - FULL TCL EXAMPLE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'bookflow_db' expanded, showing tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor shows a TCL script with the following content:

```
68 WHERE txn_id=102;  
69 UPDATE bank_transactions  
70 SET amount=7000  
71 WHERE txn_id=101;  
72  
73 SAVEPOINT SP1;  
74  
75 UPDATE bank_transactions  
76 SET amount=9000  
77 WHERE txn_id=102;  
78  
79 ROLLBACK TO SP1;  
80  
81 COMMIT;  
82
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." Below this is a 'Context Help' tab and a 'Snippets' tab.

The bottom panel shows the 'Output' window with the following data:

#	Time	Action	Message	Duration / Fetch
48	10:26:57	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
49	10:27:17	UPDATE customer_transactions SET amount=6000 WHERE txn_id=101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
50	10:27:58	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
51	10:28:36	UPDATE bank_transactions SET amount=99999 WHERE txn_id=102	Error Code: 1146. Table 'bookflow_db.bank_transactions' doesn't exist	0.000 sec
52	10:28:50	UPDATE customer_transactions SET amount=99999 WHERE txn_id=102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
53	10:29:39	COMMIT	0 row(s) affected	0.000 sec

The bottom status bar indicates "Query Completed".